

PMATH 453 Funcional Analysis

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1 Set Theory and Topology

1.1 Topology

(X, d) metric space. Then $U \subseteq X$ open $\iff U = \text{union of open balls}$. For $x \in X, r > 0$,

$$b_r(x) = \{y \in X \mid d(x, y) < r\}$$

“open ball centred at x with radius r . So

$$U \text{ open} \iff \forall x \in U, \exists r > 0 \text{ such that } b_r(x) \subseteq U.$$

Recall: $f : (X_1, d_1) \rightarrow (X_2, d_2)$ is continuous $\iff f^{-1}(\text{open set})$ is open.

Definition 1.1. $X \neq \emptyset$. A topology on X is a collection $\tau \subseteq \mathcal{P}(X) = 2^X = \text{power set of } X$, satisfying

1. $\emptyset, X \in \tau$
2. τ closed under arbitrary union
3. τ closed under finite intersection.

$(X, \tau) = \text{“topological space”}$. $U \in \tau$ is an open set. We say $F \subseteq (X, \tau)$ is closed $\iff F^c$ is open.

Example 1.2. 1. (X, d) metric space. We say τ_d is the metric topology generated by open balls.

2. X set, trivial topology $\tau = \{\emptyset, X\}$.
3. X set, discrete topology $\tau = \mathcal{P}(X)$
4. order topology: Let $(X, \leq)$ be a partially ordered set, i.e., $\leq$ is a binary operation satisfies
 - anti-symmetric: $x \leq y$ and $y \leq x \implies x = y$
 - reflexive: $x \leq x, \forall x$
 - transitive: $x \leq y, y \leq z \implies x \leq z$.

A totally ordered set is a poset (partial ordered set) $(X, \leq)$ such that $\forall x, y \in X$, we have $x \leq y$ or $y \leq x$. If $(X, \leq)$ is a totally ordered set, then for $a \in X$, we have

$$\begin{cases} (-\infty, a) = \{x \in X \mid x < a\} = \{x \in X \mid x \leq a, x \neq a\} \\ (a, \infty) = \{x \in X \mid a < x\} = \{x \in X \mid a \leq x, x \neq a\} \\ (a, b) = (a, \infty) \cap (-\infty, b) \end{cases}$$

and all unions of above to be open. This defines $\tau_{\leq} \subseteq \mathcal{P}(X)$. CHECK: $\tau_{\leq}$ is a topology.

Definition 1.3. For $A \subseteq X$, we define the closure and interior of A to be

$$\overline{A} = \bigcap_{\substack{F \supseteq A \\ \text{closed}}} F = \text{smallest closed set containing } A$$

$$A^\circ = \bigcup_{\substack{U \subseteq A \\ \text{open}}} U = \text{largest open set contained in } A$$

Proposition 1.4. Recall from previous courses,

1. closed sets are stable under finite unions, arbitrary intersections.
2. $\overline{A} = ((A^c)^\circ)^c$
3. $x \in \overline{A} \iff \forall U \text{ open, and } x \in U \implies U \cap A \neq \emptyset$

Let $S \subseteq \mathcal{P}(X)$ to be arbitrary. We can then define

τ_S = topology generated by S = smallest topology on X containing $S = \{\emptyset, X, \text{all unions of finite intersections of members of } S\}$

Bases: (X, τ) . A base for τ is a family of open sets $\{U_\alpha\}_{\alpha \in A}$ such that every $U \in \tau$ is a union of some U_α 's. A subbase is a collection of open sets $S \subseteq \tau$ such that

$$\{S_1 \cap \cdots \cap S_n \mid n \geq 1, S_i \in S\}$$

forms a base.

Example 1.5. $(X, d), \{b_{\frac{1}{n}}(x) \mid x \in X, n \in \mathbb{N}\}$.

Example 1.6. Totally ordered set $(X, \leq)$.

Definition 1.7. $(X, \tau), x \in X$. A neighborhood of x is a set $V \ni x$ such that $\exists U \in \tau$ with $x \in U \subset V$.

1.2 Countability Axioms

Definition 1.8. (X, τ)

- (0) Let $S \subset X$. Say S is dense in (X, τ) if $\forall$ open set $U \neq \emptyset, S \cap U \neq \emptyset$. (Or equivalently, $\overline{S} = X$).
- (1) Say (X, τ) is separable if there is a countable dense subset.
- (2) Say (X, τ) is first countable if $\forall x \in X, \exists$ a countable (open) neighborhood base $\{U_n\}_{n=1}^\infty \subset \tau$ at x . That is, for any neighborhood V of x , there exists $n \in \mathbb{N}$ such that $x \in U_n \subseteq V$.
- (3) (X, τ) is second countable if it has a countable base for τ .

Example 1.9. (a) Metric spaces (X, d) are first countable.

(b) $(X, \tau_{\text{discrete}})$ is separable $\iff |X| \leq |\mathbb{N}|$.

(c) (X, d) is separable $\iff$ it is second countable.

Proof. ($\implies$) If $S = \{x_k\}_{k=1}^\infty$ is dense. Then $\{b_r(x_k) \mid r \in \mathbb{Q}, k \in \mathbb{N}\}$ is a countable base.

($\impliedby$) General case. If X is second countable. Let $\{U_n\}_{n=1}^\infty$ be the base. WLOG, $U_n \neq \emptyset$. Pick (by axiom of countable choice) $\forall n \in \mathbb{N}, x_n \in U_n$. Let $S = \{x_n\}_{n=1}^\infty$. If $U \neq \emptyset$ open with $U_n \subset U$, then $x_n \in U$ so $S \cap U \neq \emptyset$. That is, S is dense. $\square$

Definition 1.10 (AOC). The Axiom of Choice states if $X \neq \emptyset$, then there exists a choice function

$$c : \mathcal{P}(X) \setminus \{\emptyset\} \rightarrow X$$

such that $\forall A \subseteq X, A \neq \emptyset$ with $c(A) \in A$.

Remark 1.11. AOC is a set theoretic statement that we take to be true.

Proposition 1.12. TFAE

1. AOC
2. AOC': Let $\{X_\alpha\}_{\alpha \in A}$ be a family of non-empty sets. Then

$$\prod_{\alpha \in A} X_\alpha := \{f : A \rightarrow \bigsqcup_{\alpha \in A} X_\alpha \mid f(\alpha) \in X_\alpha\} \neq \emptyset$$

Proof. ($\implies$) Let $X = \bigsqcup_{\alpha} X_\alpha$. Then we have choice function $c : \mathcal{P}(X) \setminus \{\emptyset\} \rightarrow X$, then

$$f : A \rightarrow \bigsqcup_{\alpha} X_\alpha \text{ via } f(\alpha) = c(X_\alpha) \in X_\alpha$$

So $f \in \prod_{\alpha} X_\alpha$

($\impliedby$) Suppose (AOC') holds, then we have

$$\prod_{X_\alpha \in \mathcal{P}(X) \setminus \{\emptyset\}} X_\alpha \neq \emptyset$$

Any c in this set is a choice function. $\square$

Definition 1.13. If $\{\tau_\alpha\}_\alpha$ is a family of topologies on X . Then there is a weakest topology that is stronger than each τ_α . There is a strongest topology σ that is weaker than each τ_α .

$$\tau = < \bigcup_{\alpha} \tau_\alpha > \quad \sigma = \bigcap_{\alpha} \tau_\alpha$$

where we say τ is weaker than τ' iff $\tau \subset \tau'$.

Definition 1.14. We say $f : (X, \tau_X) \rightarrow (Y, \tau_Y)$ is continuous if $f^{-1}(U) \in \tau_X, \forall U \in \tau_Y$.

Definition 1.15. We say $f : X \rightarrow Y$ is open if $f(V)$ is open $\forall V \subseteq X$ open.

Definition 1.16. We say $f : X \rightarrow Y$ is a homeomorphism if f is continuous, bijective and f^{-1} is continuous.

Definition 1.17. Say (X, τ) is Hausdorff (T2) if $\forall x \neq y \in X, \exists U \ni x, V \ni y$ open such that $U \cap V = \emptyset$.

Example 1.18. 1. Metric spaces are Hausdorff

2. $|X| \geq 2, (X, \tau_{\text{triv}})$ is non-Hausdorff.

3. $(X, \tau_{\text{discrete}})$ is always Hausdorff

4. $X = (0, 1) \cup \{1^+\} \cup \{1^-\}$, with $(0, 1)$ on usual topology, and declare $(r, 1) \cup \{1^\pm\}$ open, $0 < r < 1$, be a base for the topology. Then X is not Hausdorff since we cannot separate 1^+ and 1^- by disjoint open sets.

Proof. Suppose $1^+ \in U, 1^- \in V$ open, then there exists $0 < r_u, r_v < 1$ such that

$$\begin{aligned} (r_u, 1) \cup \{1^+\} &\subseteq U, (r_v, 1) \cup \{1^-\} \subseteq V \\ \implies \emptyset \neq U \cap V &\supseteq (\max(r_u, r_v), 1) \end{aligned}$$

□

Proposition 1.19. If X is Hausdorff, then singletons are closed.

Proof. Let $x \in X$. Then for all $y \neq x$, there exists a neighborhood $V_y \ni y$ open such that $V_y \cap \{x\} = \emptyset$. Then $X \setminus \{x\} = \bigcup_y V_y$ open, therefore $\{x\}$ is closed. □

Definition 1.20. We write

1. $C(X) := \{f : X \rightarrow \mathbb{C} \mid f \text{ is continuous}\}$
2. $C(X, \mathbb{R}) := \{f : X \rightarrow \mathbb{R} \mid f \text{ is continuous}\}$
3. $C_b(X) := \{f \in C(X) \mid \|f\|_\infty = \sup_{x \in X} |f(x)| < \infty\}$
4. $C_b(X, \mathbb{R}) := \{f \in C(X, \mathbb{R}) \mid \|f\|_\infty = \sup_{x \in X} |f(x)| < \infty\}$

Remark 1.21. $C(X), C_b(X), \dots$ are unital, commutative, associate algebras over $\mathbb{C}$, and $\|\cdot\|_\infty$ is a norm on $C_b(X)$.

Question: When does $C(x)$ (or $C_b(X)$) contain “lots of ” functions? Can $C(X)$ (or $C_b(X)$) separates points of X ?

Example 1.22. $(X, \tau_{\text{triv}}), |X| \geq 2$. One can check that $C(X) = \mathbb{C} \times 1$, where 1 is the constant 1 function, really small.

(X, τ_{disc}) , then $C(X)$ separates points.

Proposition 1.23. If $C(X)$ separates points, then so does $C_b(X), C_b(X, \mathbb{R})$, and X is Hausdorff.

Proof. If $x \neq y$, $f \in C(X)$, $f(x) \neq f(y)$. WLOG, $\text{Ref}(x) \neq \text{Ref}(y)$. Assume f is real-valued. WLOG, $f(x) < f(y)$. Let $g = \min\{f(y), \max\{f, f(x)\}\}$, then g is continuous, and

$$f(x) \leq g \leq f(y), g(x) = f(x), g(y) = f(y)$$

If $x \neq y$, $\exists f \in C(X)$, say $|f(x) - f(y)| = r > 0$. Then

$$x \in U = f^{-1}(b_{\frac{r}{2}}(f(x))), y \in V = f^{-1}(b_{\frac{r}{2}}(f(y)))$$

Then $U \cap V = f^{-1}(b_{\frac{r}{2}}(f(x)) \cap b_{\frac{r}{2}}(f(y))) = f^{-1}(\emptyset) = \emptyset$ □

Definition 1.24. (X, τ) is normal if for all A, B closed with $A \cap B = \emptyset$, then there exists $A \subseteq U, B \subseteq V$ open such that $U \cap V = \emptyset$.

Theorem 1.25 (Urysohn's Lemma). X is normal $\iff$ A, B closed with $A \cap B = \emptyset$, then there exists $f : X \rightarrow [0, 1]$ continuous such that $f|_A = 0, f|_B = 1$.

Corollary 1.26. Normal Hausdorff $X \implies C_b(X)$ separates points.

Example 1.27. (X, d) metric space $\implies$ normal. Take A, B closed, $A \cap B = \emptyset$, then we can let

$$f(x) = \frac{\text{dist}(x, A)}{\text{dist}(x, A) + \text{dist}(x, B)}.$$

Example 1.28. Let $X = \mathbb{N}_0 \times \mathbb{N}_0 = \{(m, n) : m, n \in \mathbb{N}_0 = \mathbb{N} \cup \{0\}\}$. Then we can define a topology on X :

1. If $U \subseteq X$, $(0, 0) \notin U$, then U is open. Let

$$\pi_1 : X \rightarrow \mathbb{N}_0 \text{ via } \pi_1(m, n) = m$$

2. If $(0, 0) \in U$, then we say U is open iff

$$\{m \in \mathbb{N}_0 \mid \pi_1^{-1}(m) \cap U \text{ cofinite in } \{m\} \times \mathbb{N}_0\} \text{ is cofinite}$$

That is, we say $U \ni (0, 0)$ is open if all but finitely many columns of U contains all but finitely many elements of the column.

Exercise: this gives a topology.

Exercise: X is Hausdorff.

Proof. Denote $X_0 = X \setminus \{(0, 0)\}$ is discrete. For $(m, n) \neq (m', n') \in X_0$, then let $U = \{(m, n)\}, V = \{(m', n')\}$. Since $(m, n), (m', n') \neq (0, 0)$, we have U, V open, and $U \cap V = \emptyset$.

If $(m, n) \neq (m', n') = (0, 0)$, then $U = \{(m, n)\}$ open, and $V = X \setminus \{(m, n)\} \ni (0, 0)$ open, and $U \cap V = \emptyset$. □

Exercise: $\overline{X_0} = X$. That is, $(0, 0) \in \overline{X_0}$

Proof. If $(0, 0) \in U$ open, clearly $U \cap X_0 \neq \emptyset$. By definition, $(0, 0) \in \overline{X_0}$. □

Exercise: $\beta(x_k)$ sequence in X_0 such that $x_k \rightarrow (0, 0)$. Notice that $x_k \rightarrow (0, 0)$ in topological space means $x_k \rightarrow x \in (X, \mathcal{T})$ if $\forall x \in U \subseteq X$ open, there exists $k_0 \in \mathbb{N}$ such that $k \geq k_0 \implies x_k \in U$.

Proof. Let $(x_k) \subseteq X_0$ be arbitrary. Say $x_k = (m_k, n_k)$.

Case 1: $m_k \leq M, \forall k$. Then take $U \ni (0, 0) = \{(m, n) \mid m > M\} \cup \{(0, 0)\}$. Therefore $x_k \notin U, \forall k$.

Case 2: there exists $k_1 < k_2 < \dots$ such that $m_{k_1} < m_{k_2} < \dots$. Then we can choose

$$U = X \setminus \{x_{k_i} \mid i \geq 1\}$$

Then it is open, and $x_{k_i} \notin U, \forall i$. Therefore $x_k \not\rightarrow (0, 0)$.

□

Definition 1.29. We define

1. An upwards directed set is a nonempty poset $(\Lambda, \leq)$ such that if $\lambda_1, \lambda_2 \in \Lambda$, there exists λ_0 such that $\lambda_1 \leq \lambda_0$ and $\lambda_2 \leq \lambda_0$.
2. Let $X = \emptyset$. A net on X is a function $j : \Lambda \rightarrow X$, where $(\Lambda, \leq)$ is an upwards directed set. Write $x_\lambda = j(\lambda) \in X$. Net $(j, \Lambda, \leq, X) \iff (x_\lambda)_{\lambda \in \Lambda}$ net in X .
3. Let (X, τ) be a topological space. Say a net $(x_\lambda)_{\lambda \in \Lambda}$ in X converges to $x \in X$ if $\forall U \ni x$ open, there exists $\lambda_0 \in \Lambda$ such that $x_\lambda \in U, \forall \lambda \geq \lambda_0$. Then we can get (y_γ) where $y_\gamma = x_{\varphi(\gamma)}$. Say $(y_\gamma)_{\gamma \in \Gamma} = (x_{\varphi(\gamma)})_{\gamma \in \Gamma}$ is a net in X .
4. We say a map $\varphi : \Gamma \rightarrow \Lambda$ is cofinal if $\forall \lambda_0 \in \Lambda$, there exists $\gamma_0 \in \Gamma$ such that $\forall \gamma \geq \gamma_0$, we have $\varphi(\gamma) \geq \lambda_0$.
5. Given a net (x_λ) in X , a subnet of (x_λ) is given by an upwards directed set $(\Gamma, \leq)$ and a function $\varphi : \Gamma \rightarrow \Lambda$ that is cofinal.
6. (X, τ) topological space. Let $x \in X$. Let

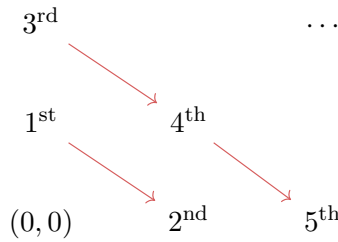
$$\mathcal{O}(x) := \{U \in \tau : x \in U\}$$

the system of open neighborhoods of x .

Example 1.30. $X = \mathbb{N}_0 \times \mathbb{N}_0$, $X_0 = X \setminus \{0\}$. Then any $U \subseteq X_0$ is open. $U \ni (0, 0)$ is open $\iff$ cofinitely many cofinite columns in U .

Claim: There exists a net in X_0 converging to $(0, 0)$.

Proof. We will prove it by enumerate X_0 .



Then we have $X_0 = \{x_k\}_{k \in \mathbb{N}}$. Let $\Lambda = \mathcal{O}((0, 0))$ upwards directed by containment

$$U \leq V \iff V \subseteq U$$

For all $U \in \Lambda$, pick $x_U = x_{K(U)}$, where $K(U)$ is the first $k \in \mathbb{N}$ such that $x_k \in U$ (Avoids AOC). $\square$

Claim: $(x_U)_{U \in \Lambda}$ converges to $(0, 0)$.

Proof. Pick $U_0 \ni (0, 0)$ for all open $U \subseteq U_0$, $x_U \in U \subseteq U_0$. Observe that the net (x_U) that we constructed is a (convergent) subnet of $(x_k)_{k \in \mathbb{N}}$. Indeed, $\Lambda = (\mathcal{O}((0, 0)), \leq)$, $(\mathbb{N}, \leq)$, $\varphi : \Lambda \rightarrow \mathbb{N}$ via $\varphi(U) = K(U)$. Also, φ cofinal since if we fix $k_0 \in \mathbb{N}$, we can find $U_0 \in \Lambda$ such that $\forall U \geq U_0$, $\varphi(U) \geq k_0$. Take $U = X \setminus \{x_1, \dots, x_{k_0}\}$, then $K(U) \geq k_0$ $\square$

Theorem 1.31. $(X, \tau_X), (Y, \tau_Y)$. Then

1. If $A \subseteq X$, then $x \in \overline{A} \iff$ there exists a net $(x_\lambda)_{\lambda \in \Lambda} \subseteq A$ such that $x_\lambda \rightarrow x$.
2. A function $f : X \rightarrow Y$ is continuous $\iff \forall$ net $(x_\lambda)_{\lambda \in \Lambda} \subseteq X$ with $x_\lambda \rightarrow x$, then $f(x_\lambda) \rightarrow f(x)$.

Proof. 1. $(\implies)$ $x \in \overline{A}$, then $\forall U \ni x$ open, $U \cap A \neq \emptyset$. By AOC, we get $U \mapsto x_U \in U \cap A$, get a net $(x_U)_{U \in \mathcal{O}(x)}$ and $x_U \rightarrow x$.
 $(\impliedby)$ If $x_\lambda \in A$ such that $x_\lambda \rightarrow x$. Then for any $U \in \mathcal{O}(x)$, there exists λ_0 such that $\lambda_0 \leq \lambda$. Then

$$x_\lambda \in U \implies U \cap A \neq \emptyset \implies x \in \overline{A}$$

2. $(\implies)$ f cts, suppose $x_\lambda \rightarrow x$, let $V \ni \mathcal{O}(f(x))$. Say $U = f^{-1}(V)$ open, then there exists λ_0 such that $\lambda \geq \lambda_0 \implies x_\lambda \in U$. Therefore $f(x_\lambda) \in f(U) = V \implies f(x_\lambda) \rightarrow f(x)$.
 $(\impliedby)$ Suppose f not continuous, then there exists $V \subseteq Y$ open with $U := f^{-1}(V)$ not open, therefore U^c is not closed. Hence $U^c \subsetneq \overline{U^c} \implies U \cap \overline{U^c} \neq \emptyset$. Then there exists $x \in U \cap \overline{U^c}$. By part 1, there exists $(x_\lambda)_{\lambda \in \Lambda} \in U^c$ such that $x = \lim_{\lambda \in \Lambda} x_\lambda$ (i.e., $x_\lambda \rightarrow x$), then

$$f(x_\lambda) \in f(U^c) = f(f^{-1}(V)^c) = f(f^{-1}(V^c)) = V^c$$

Contradicts to $f(x_\lambda) \rightarrow f(x)$. $\square$

Definition 1.32. (X, τ) . Then a open cover is a collection $\{U_\alpha\}_{\alpha \in A}$ open sets such that

$$X = \bigcup_{\alpha \in A} U_\alpha.$$

A finite subcover is $F \subseteq A$ finite such that

$$X = \bigcup_{\alpha \in F} U_\alpha.$$

Definition 1.33. (X, τ) is compact if every open cover of X has a finite subcover.

Definition 1.34. A collection $\{C_\alpha\}_{\alpha \in A}$ of non-empty closed sets is a (FIP)-family if for all $F \subseteq A$ finite,

$$\bigcap_{\alpha \in F} C_\alpha \neq \emptyset$$

We say (X, τ) has the finite intersection property (FIP) if for all (FIP)-families $\{C_\alpha\}_\alpha$, we have

$$\bigcap_{\alpha \in A} C_\alpha \neq \emptyset$$

Theorem 1.35. (X, τ) . TFAE:

1. X is compact,
2. X has the FIP,
3. for all nets $(x_\lambda) \subseteq X$, there exists a convergent subnet.

Proof. (1) $\implies$ (2): Suppose, for contradiction, that $\{C_\alpha\}$ is a FIP family and $\bigcap_{\alpha \in A} C_\alpha = \emptyset$, then $X = \bigcup_\alpha C_\alpha^c$, an open cover. By compactness, there exists $F \subseteq A$ finite with $X = \bigcup_{\alpha \in F} C_\alpha^c$, therefore

$$\emptyset = \bigcap_{\alpha \in F} C_\alpha$$

contradicts to $\{C_\alpha\}$ is a FIP family.

(2) $\implies$ (1): Let $X = \bigcup_{\alpha \in A} U_\alpha$ be an arbitrary open cover, then

$$\emptyset = \bigcap_{\alpha \in A} U_\alpha^c$$

by assumption of X has the FIP, there must exists some $F \subseteq A$ finite such that

$$\emptyset = \bigcap_{\alpha \in F} U_\alpha^c$$

Then $X = \bigcup_{\alpha \in F} U_\alpha$ is a finite subcover.

(2) $\implies$ (3): Let $(x_\lambda)_{\lambda \in \Lambda}$ be a net in X . Then

$$C_\lambda := \overline{\{x_\mu \mid \mu \geq \lambda\}} \neq \emptyset$$

Claim: $\{C_\lambda\}_{\lambda \in \Lambda}$ is a FIP family.

Fix $\lambda_1, \dots, \lambda_n \in \Lambda$, then there exists $\lambda_0 \geq \lambda_i, \forall i = 1, \dots, n$. Therefore

$$C_{\lambda_1} \cap C_{\lambda_2} \cap \dots \cap C_{\lambda_n} \supseteq C_{\lambda_0} \ni x_{\lambda_0}$$

which finishes the proof of claim. By assumption of X has the FIP, we have

$$\bigcap_{\lambda \in \Lambda} C_\lambda \neq \emptyset$$

Say $x \in \bigcap_{\lambda \in \Lambda} C_\lambda$. We now construct a subnet converging to x :

Let $\Gamma := \Lambda \times \mathcal{O}(x)$, and we define $(\lambda, U) \leq (\lambda', U')$ if

1. $\lambda \leq \lambda'$,
2. $U' \subseteq U$.

Fix $(\lambda, U) \in \Gamma$. Since $x \in C_\lambda = \overline{\{x_\mu \mid \mu \geq \lambda\}}$, we have $U \cap \{x_\mu \mid \mu \geq \lambda\} \neq \emptyset$. By AOC,

$$\exists y_{(\lambda, U)} = x_{\mu(\lambda, U)} \in U \cap \{x_\mu \mid \mu \geq \lambda\}$$

Then we get $\varphi : \Gamma \rightarrow \Lambda$ via $\varphi(\lambda, \mu) = \mu(\lambda, U)$, and $(y_{(\lambda, U)})$ is a subnet, and

$$y_{(\lambda, U)} \rightarrow x.$$

φ cofinal: Fix $\lambda_0 \in \Lambda$. Let $\gamma_0 = (\lambda_0, X)$. If $\gamma = (\lambda, U) \geq \gamma_0$, then

$$\lambda \geq \lambda_0, U \subseteq X$$

and then $\varphi(\gamma) = \mu(\lambda, U) \geq \lambda \geq \lambda_0$. Therefore, for all $U_0 \in \mathcal{O}(x)$, for all $\lambda_0 \in \Lambda$, $(\lambda, U) \geq (\lambda_0, U_0) \implies y_{(\lambda, U)} = x_{\varphi(\lambda, U)} \in U \subseteq U_0$.

(3) $\implies$ (2): Fix a FIP family $\{C_\alpha\}_{\alpha \in A}$ in X . For any $F \subseteq A$ finite, there exists $x_F \in \bigcap_{\alpha \in F} C_\alpha$. We get a net $(x_F)_{F \subseteq A, \text{ finite}}$. Say $F_1 \leq F_2$ if $F_1 \subseteq F_2 \subseteq A$. Then $F_1, F_2 \leq F_1 \cup F_2$, therefore $(\{F\}, \leq)$ it is a upwards directed set. By assumption of (3), there exists a convergent subnet

$$\varphi : \Gamma \rightarrow \Lambda \quad \text{cofinal}$$

and $(x_{\varphi(\gamma)})_{\gamma \in \Gamma} \rightarrow x \in X$.

Claim: $x \in \bigcap_{\alpha \in A} C_\alpha$.

Fix $\alpha \in A$, $\lambda_0 := \{\alpha\} \in \Lambda$. φ cofinal $\implies$ there exists $\gamma_0 \in \Gamma$ such that

$$\varphi(\gamma) \geq \lambda_0 \implies \alpha \in \varphi(\gamma), \forall \gamma \geq \gamma_0$$

Therefore, $x_{\varphi(\gamma)} \in \bigcap_{\beta \in \varphi(\gamma)} C_\beta \subseteq C_\alpha, \forall \gamma \geq \gamma_0$ since $\alpha \in \varphi(\gamma)$. Then

$$x \in \overline{\{x_{\varphi(\gamma)} \mid \gamma \geq \gamma_0\}} \subseteq \overline{C_\alpha} = C_\alpha$$

Since α is chosen arbitrarily, we finish the proof of the claim. □

Proposition 1.36. If X is compact, and Y is a topological space, $f : X \rightarrow Y$ is continuous. Then $f(X)$ is compact in Y .

Proof. Exercise. □

Proposition 1.37. If X is compact, then:

1. If $A \subseteq X$ closed, then A is compact.
2. If X is Hausdorff, then $A \subseteq X$ compact $\implies A$ is closed.

Proof. 1. Exercise.

2. Suppose A is compact. Then for any $x \in A^c$, for any $y \in A$, there exists U_x, V_x open such that $x \in U_x, y \in V_x, U_x \cap V_x = \emptyset$. Therefore, there exists $V_{x_1} \cup \dots \cup V_{x_n} \supseteq A$.

Claim: $x \in \bigcap_{i=1}^n U_{x_i} \subseteq A^c$ Therefore, A^c is open.

□

[make up here]

Proposition 1.38. X LCH. If $K \subseteq X$ compact, say $K \subseteq \bigcup_{i=1}^n U_i$, then there exists $h_1, \dots, h_n \in C_c(X)$ such that

- $0 \leq h_i \leq 1$
- $h_i|_{U_i^c} = 0$
- $h_1 + h_2 + \dots + h_n = 1$ on K .

Definition 1.39. We define

$$C_0(X) := \overline{C_c(X)}^{\|\cdot\|_\infty} \subseteq C_b(X)$$

the continuous function vanishing at ∞ . More formally, $f \in C_0(X) \iff \forall \varepsilon > 0, \exists K \subseteq X$ compact such that $|f(x)| < \varepsilon$ on $X \setminus K$. Then $C_0(X)$

- Banach space w.r.t. $\|\cdot\|_\infty$
- Algebra structure
- involution $f \mapsto f^*$, where $f^*(x) = \overline{f(x)}$
- $(C_0(X), \|\cdot\|_\infty)$ is a commutative C^* -algebra.

Theorem 1.40 (Gelfand (Naimark)). Any commutative C^* -algebra $(A, \|\cdot\|)$ is $(*)$ -isomorphic to $C_0(X)$ for some unique LCH X , where $(*)$ -isomorphism is unital algebra isomorphism and preserves involution.

Exercise: $\pi : C_0(X) \rightarrow C_0(Y)$ $*$ -isomorphism via $\pi(f)^* = \pi(f^*)$. Then there exists unique $\Phi : Y \rightarrow X$ such that

$$\pi(f) = f \circ \Phi$$

1.3 Product Topology

Definition 1.41. Let (X_α, τ_α) be topological spaces. Then we topologize $\prod_{\alpha \in A} X_\alpha$. We define the product topology to be generated by the sets

$$\{U_\alpha \times \prod_{\beta \in A \setminus \{\alpha\}} X_\beta \mid U_\alpha \in \tau_\alpha\} = \{\pi_\alpha^{-1}(U_\alpha) \mid \alpha \in A, U_\alpha \in \tau_\alpha\}$$

where we set

$$\pi_\alpha : \prod_{\beta \in A} X_\beta \rightarrow X_\alpha \quad \text{via} \quad (x_\beta)_{\beta \in A} \rightarrow x_\alpha$$

the “projection map onto X_α ”.

Remark 1.42. Product topology is the weakest topology making each $\pi_\alpha : \prod_\beta X_\beta \rightarrow X_\alpha$ continuous.

Proposition 1.43. A net $(x_\lambda)_{\lambda \in \Lambda}$ in $\prod_{\alpha \in A} X_\alpha$ converges to $x \iff x_\lambda(\alpha) \rightarrow x(\alpha) \in X_\alpha, \forall \alpha \in A$, where $x_\lambda = (x_\lambda(\alpha))_{\alpha \in A}$ and $x = (x(\alpha))_{\alpha \in A}$.

Proof. Exercise! □

Key idea: base for product topology:

$$\pi_{\alpha_1}^{-1}(U_{\alpha_1}) \cap \cdots \cap \pi_{\alpha_n}^{-1}(U_{\alpha_n}) = U_{\alpha_1} \times \cdots \times U_{\alpha_n} \times \prod_{\beta \neq \alpha_i} X_\beta$$

where $U_{\alpha_i} \in \tau_{\alpha_i}$.

Theorem 1.44. TFAE:

1. Tychonoff Theorem
2. AOC
3. Zorn's Lemma
4. Well ordering principle.

X compact Hausdorff space. Then $(C(X), \|\cdot\|_\infty)$ is a Banach space.
 X LCH, then $(C_b(X), \|\cdot\|_\infty), (C_0(X), \|\cdot\|_\infty)$ are Banach spaces.

Notation:

$c_0 = C_0(\mathbb{N}) = \{(x_n)_{n=1}^\infty : \lim_{n \rightarrow \infty} x_n = 0\}$, where $\mathbb{N}$ is LCH with discrete topology.

$c_0(\mathbb{R}) = \{f : \mathbb{R} \rightarrow \mathbb{F} : f \text{ continuous, and } \lim_{n \rightarrow \pm\infty} f(x) = 0\}$

Example 1.45. $C^n[0, 1] = \{f : [0, 1] \rightarrow \mathbb{F} : f^{(n)} \text{ continuous}\}$. We can define

$$\|f\|_{C^n[0,1]} = \sum_{i=1}^n \|f^{(i)}\|_\infty,$$

this is a norm, and $C^n[0, 1]$ is complete with respect to this norm. Therefore, $C^n[0, 1]$ is a Banach space.

Proof. For f_k Cauchy in $\|\cdot\|_{C^n[0,1]}$, then $(f_k^{(i)})_k$ is Cauchy in $\|\cdot\|_\infty$ for all $i = 0, \dots, n$. Thus

$$\begin{aligned} & \xRightarrow[\text{unif cts}]{f_k^{(i)}} g_i \in C[0, 1] \\ & \xRightarrow{\text{FTC}} f_k^{(i)}(x) = f_k^{(i)}(0) + \int_0^x f_k^{(i+1)}(t) dt \\ & \implies g_i(x) = g_i(0) + \int_0^x g_{i+1}(t) dt \end{aligned}$$

Therefore g'_i exists and equals g_{i+1} . □

2 Measure Theory

$(X, \mathcal{M}, \mu)$ measure space.

Example 2.1. $X = \mathbb{R}$ or $[0, 1]$. $\mu : \mathcal{M} \rightarrow [0, \infty]$ = Lebesgue measure. $\mathcal{M}$ = Lebesgue measure sets or Borel sets. For $1 \leq p < \infty$, $p \in \mathbb{R}$. Then

$$L^p(\mu) = L^p(X) = L^p(X, \mu) = L^p = \{f : X \rightarrow \mathbb{C} : f \text{ measurable and } \int_X |f|^p d\mu < \infty\}$$

and

$$\|f\|_p = \left(\int_X |f|^p d\mu \right)^{1/p}$$

Also, when $p = \infty$, we define

$$L^\infty(\mu) = \{f : X \rightarrow \mathbb{C} : f \text{ measurable such that } \|f\|_{L^\infty(\mu)} < \infty\}$$

where

$$\|f\|_{L^\infty(\mu)} = \inf\{A > 0 : \mu(\{x \in X : |f(x)| > A\}) = 0\}$$

and we write $\|\cdot\|_\infty$ for the norm on L^∞ .

Remark 2.2. On $L^p(\mu)$, identify $f, g \in L^p$ are equal if $f - g = 0$ ae.

Proposition 2.3 (Hölder's inequality). Let $1 \leq p < \infty$, $q \in (1, \infty]$ Hölder conjugate ($\frac{1}{p} + \frac{1}{q} = 1$). Let $f \in L^p, g \in L^q$, then $fg \in L^1$, and

$$\|fg\|_1 = \int_X |fg| d\mu \leq \|f\|_p \|g\|_q$$

By Hölder's inequality, we obtain triangle inequality, therefore $(L^p(\mu), \|\cdot\|_p)$ is a normed vector space. Dominated convergence theorem implies $L^p(\mu)$ is a Banach space.

Example 2.4. $X = \mathbb{N}$, $\mathcal{M} = 2^{\mathbb{N}}$, μ = counting measure. Then

$$L^p(\mu) = l^p = l_p = \{(x_i)_{i=1}^\infty : \|x\|_p = \left(\sum_{i=1}^\infty |x_i|^p \right)^{1/p} < \infty\}$$

$$L^\infty(\mu) = l^\infty = l_\infty = \{(x_i)_{i=1}^\infty : \|x\|_\infty = \sup_i |x_i| < \infty\}$$

2.1 Hilbert Space

H = complete inner product space over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. Then $\langle \cdot, \cdot \rangle : H \times H \rightarrow \mathbb{F}$ satisfies

1. $\langle x, x \rangle \geq 0$ positivity
2. $\langle x, x \rangle = 0 \iff x = 0$

3. sesquilinearity: linear in the first variable and conjugate linear in the second variable

$$\langle \alpha x + y, z \rangle = \alpha \langle x, z \rangle + \langle y, z \rangle$$

$$\langle x, y \rangle = \overline{\langle y, x \rangle}$$

$$\langle z, \alpha x + y \rangle = \overline{\alpha} \langle z, x \rangle + \langle z, y \rangle$$

4. on H , we write $\|x\| = (\langle x, x \rangle)^{\frac{1}{2}}$

Proposition 2.5 (Cauchy Schwarz). $|\langle x, y \rangle| \leq \|x\| \|y\|$

Proof. $x, y \in H$. WLOG, $y \neq 0$. Fix $t \in \mathbb{F}$. Then

$$\begin{aligned} 0 &\leq \|x + ty\|^2 \\ &= \langle x + ty, x + ty \rangle \\ &= \|x\|^2 + 2\operatorname{Re}(\bar{t}\langle x, y \rangle) + |t|^2 \|y\|^2, \forall t \end{aligned}$$

Pick $t = -\frac{\langle x, y \rangle}{\|y\|^2}$. □

Definition 2.6. (X, Y) NVS, $T \in B(X, Y)$.

We call T a contraction if

$$\|T\| \leq 1$$

We say T is an isometry if

$$\|Tx\| = \|x\|, \forall x \in X$$

We say T is bounded below if there exists $c > 0$ such that

$$\|Tx\| \geq c\|x\|, \forall x \in X$$

We say T is an isomorphism if T is surjective and bounded below. Then $T^{-1} \in B(Y, X)$, with $\|T^{-1}\| \leq c$ when T is bounded below by c .

We say T is an isometric isomorphism $\iff T$ is a surjective isometry.

Remark 2.7. $C_0^* \simeq l_1$ is an isometric isomorphism. That is, there exists a surjective (linear) isometry

$$\Phi : C_0^* \rightarrow l_1$$

Last time: Given $\varphi \in C_0^*$, then let $a_\varphi = (a_1, \dots, a_n, \dots) \in l_1$, where $a_n = \varphi(e_n)$. We get

$$\|a_\varphi\|_{l_1} \leq \|\varphi\|_{C_0^*}.$$

We get a linear contraction

$$\Phi : C_0^* \rightarrow l_1, \Phi(\varphi) = a_\varphi$$

Note: Φ is surjective: if $a \in l_1$, then we can set $\varphi_a \in C_0^*$ to be

$$\varphi_a(x) = \sum_{n=1}^{\infty} x_n a_n$$

and

$$\begin{aligned} |\varphi_a(x)| &\leq \sum_{n=1}^{\infty} |x_n| |a_n| \leq \|x\|_{\infty} \|a\|_1 \\ &\implies \|\varphi_a\|_{C_0^*} \leq \|a\|_1 \end{aligned}$$

then $\Phi(\varphi) = a$, surjective. Hence

$$\begin{aligned} \|\Phi(\varphi)\| &= \|a_{\varphi}\| \leq \|\varphi\| = \|\varphi_{a_{\varphi}}\| \leq \|a_{\varphi}\| \\ &\implies \|\Phi(\varphi)\| = \|\varphi\| \end{aligned}$$

We now know $C_0^* = B(C_0, \mathbb{F})$, can we describe $B(C_0)$? $T \in B(C_0) = B(C_0, C_0)$. Associate to T , $T \rightarrow (t_{ij})_{i,j \in \mathbb{N}}$, a $\infty \times \infty$ matrix, with

$$Te_n = \begin{pmatrix} t_{1n} \\ t_{2n} \\ t_{3n} \\ \vdots \end{pmatrix} \in C_0$$

where the e_n here is the column vector, and we let δ_n to be the row vector with n th position equals to 1 and 0 everywhere else. Let

$$Tx = \begin{pmatrix} \vdots \\ \varphi_i(x) \\ \vdots \end{pmatrix} \leftarrow i^{\text{th}}$$

Then for all i , we get functional $\varphi_i \in C_0^*$, $\varphi_i(x) = \delta_i \circ T(x)$, where δ_i is the function to take i th position. Then

$$\|\varphi_i\| = \|\delta_i \circ T\| \leq \|\delta_i\| \|T\| \leq \|T\|, \forall i$$

Fix i , then $(t_{ij})_{j=1}^{\infty} = (\varphi_i(e_1), \varphi_i(e_2), \dots)$, rows of matrix (t_{ij}) all l_1 sequences. So we get $T \rightarrow$ array (t_{ij}) such that

- columns are in C_0
- rows are in l_1

with $\sup_i \|(t_{ij})_{j=1}^{\infty}\|_1 = \|\varphi_i\| \leq \|T\|$. Note

$$\begin{aligned} \|Tx\| &= \left\| \begin{pmatrix} \varphi_1(x) \\ \varphi_2(x) \\ \vdots \end{pmatrix} \right\| = \sup_i |\varphi_i(x)| \leq \sup_i \|\varphi_i\| \|x\| \\ &\implies \|T\| \leq \sup_i \|\varphi_i\| \end{aligned}$$

Therefore, $\|T\| = \sup_i \|\varphi_i\|$.

Conversely, suppose, given $(t_{ij})_{i,j \in \mathbb{N}}$ with

- columns in C_0
- rows in l_1 , uniformly bounded

can we get $(t_{ij}) \rightarrow T \in B(C_0)$? Define:

$$Tx = \begin{pmatrix} \varphi_1(x) \\ \varphi_2(x) \\ \vdots \end{pmatrix} \in l_\infty$$

where $\varphi_i(x) = \sum_{j=1}^{\infty} x_j t_{ij}$. Then

$$\|Tx\|_\infty = \sup_i |\varphi_i(x)| \leq C\|x\|$$

where C is the upper bound for l_1 norm on rows. Then T is linear. We get $T \in B(C_0, l_\infty)$. Actually, $T_{C_0} \subseteq C_0$, so $T \in B(C_0)$.

3 Functional

3.1 Subspaces and Quotients

$C(\mathbb{T})$ = continuous functions on $\mathbb{T} = \{z : |z| = 1\}$. For $n \in \mathbb{Z}$, the n th Fourier coefficient is

$$\hat{f}_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(e^{it}) e^{int} dt = \varphi_n(f)$$

where $\varphi_n \in C(\mathbb{T})^*$ given by integration.

Definition 3.1 (Disk Algebra). $A(\mathbb{D}) = \{f \in C(\mathbb{T}) \mid \hat{f}_n = 0, \forall n < 0\} = \bigcap_{n < 0} \ker(\varphi_n)$ is a closed subspace.

Why disk algebra? $\mathbb{D} = \{|z| < 1\}$, $A(\mathbb{D})$ subalgebra of $C(\mathbb{T})$.

Proof. $f, g \in C(\mathbb{T})$, then

$$\hat{f}g(n) = \sum_{k \in \mathbb{Z}} \hat{f}(k) \hat{g}(n - k)$$

If $\hat{f}(n), \hat{g}(n) = 0$ for $n < 0$, then

$$\hat{f}g(n) = \sum_{k \geq 0} \hat{f}(k) \hat{g}(n - k) = 0, n < 0$$

□

Remark 3.2. $A(\mathbb{D})$ is exactly those $f \in C(\mathbb{T})$ admitting an extension $F \in C(\overline{\mathbb{D}})$ with $F|_{\mathbb{D}}$ analytic. Obvious extension:

$$F(z) = \sum_{n=0}^{\infty} \hat{f}(n) z^n$$

Why can we take radial limits $z \in \mathbb{D} \rightarrow z_0 \in \mathbb{T}$?

Definition 3.3 (Poisson Kernal). $0 < r < 1$, we define

$$P_r(e^{it}) = \sum_{n=-\infty}^{\infty} r^{|n|} e^{int}$$

so we get $\frac{1}{2\pi} \int_{-\pi}^{\pi} P_r = 1, \forall r$. We have $P_0 = 1, P_2 = \delta_{\{0\}}, \dots$ Anyways, $0 \leq r \leq 1$, we can derive

$$\begin{aligned} F(re^{i\theta}) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(e^{it}) P_r(e^{i(\theta-t)}) dt \\ &= (f * P_r)(e^{i\theta}) \\ &= \sum_{n \in \mathbb{Z}} \hat{f}(n) r^{|n|} e^{in\theta} \\ &= \sum_{n \geq 0} \hat{f}(n) (re^{i\theta})^n \\ &= \sum_{n \geq 0} \hat{f}(n) z^n, \quad z = re^{i\theta} \end{aligned}$$

In fact, $f = \lim_{r \rightarrow 1} f * P_r$ uniformly.

$$f(e^{i\theta}) - (f * P_r)(e^{i\theta}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} (f(e^{i\theta}) - f(e^{i(\theta-t)})) P_r(e^{it}) dt$$

3.2 Quotient spaces

Let X be a Banach space and $Y \subseteq X$ be a closed subspace. Then

$$X/Y = \{x + Y : x \in X\}$$

Notation:

$$\hat{x} = x + Y = \{x + y : y \in Y\}$$

$$Q : X \rightarrow X/Y, Q(x) = \hat{x}$$

X/Y is a vector space, with

1. $(x + Y) + (z + Y) = (x + z) + Y$
2. $\alpha(x + Y) = \alpha x + Y$
3. $\hat{0} = Y$

Definition 3.4. $\|\cdot\|_{X/Y} : X/Y \rightarrow [0, \infty)$ via

$$\|\hat{x}\|_{X/Y} = \|x + Y\|_{X/Y} = \inf_{y \in Y} \|x + y\|_X$$

Proposition 3.5. $(X/Y, \|\cdot\|_{X/Y})$ is a Banach space.

Proof. $\|\hat{x}\|_{X/Y} = 0 \iff \inf_{y \in Y} \|x + y\|_X = 0 \iff x \in \overline{Y} = Y \iff \hat{x} = 0$.
Scaling is easy to check, we here prove the triangle inequality:

$$\begin{aligned} \|\hat{x} + \hat{z}\|_{X/Y} &= \inf_{y \in Y} \|x + z + y\|_X \\ &= \inf_{y_1, y_2 \in Y} \|x + y_1 + z + y_2\|_X \\ &\leq \inf_{y_1, y_2 \in Y} \|x + y_1\|_X + \|z + y_2\|_X \\ &= \|\hat{x}\|_{X/Y} + \|\hat{z}\|_{X/Y} \end{aligned}$$

It suffices to prove the completeness now: let $(\hat{x}_n)_{n=1}^\infty \subseteq X/Y$ be Cauchy. First, we can find $(\hat{x}_{n_i})_{i=1}^\infty$ such that

$$\|\hat{x}_{n_{i+1}} - \hat{x}_{n_i}\|_{X/Y} < 2^{-i}$$

Pick $z_{n_1} \in X$ such that $\hat{z}_{n_1} = \hat{x}_{n_1}$. Now

$$\|\hat{x}_{n_2} - \hat{x}_{n_1}\|_{X/Y} = \inf_{y \in Y} \|x_{n_2} - z_{n_1} + y\|_X < \frac{1}{2}$$

Then there exists y such that $z_{n_2} = x_{n_2} + y$. Then

$$\|z_{n_2} - z_{n_1}\|_X < \frac{1}{2}$$

Inductively, we find $z_{n_3}, z_{n_4}, \dots$ such that

$$\|z_{n_{i+1}} - z_{n_i}\| < 2^{-i}, \hat{z}_{n_i} = \hat{x}_{n_i}, \forall i$$

Then $(z_{n_i}) \subseteq X$ is Cauchy, so $z_{n_i} \rightarrow z \in X$. Put $\hat{z} = Q(z) \in X/Y$.

Claim: $\hat{x}_n \rightarrow \hat{z}$.

Proof. We have

$$\|\hat{x}_{n_i} - \hat{z}\|_{X/Y} = \|Q(z_{n_i} - z)\|_{X/Y} \leq \|z_{n_i} - z\|_X \rightarrow 0$$

□

Finish, $\hat{x}_n$ Cauchy convergent subsequence. □

Remark 3.6. Notice that $Q : X \rightarrow X/Y$ contraction. $\|Q\| < 1$? Yes! If $X = Y$, then $\|Q\| = 0$. If $Y \subsetneq X$, then $\|Q\| = 1$.

Example 3.7. X compact Hausdorff. $(C(X), \|\cdot\|_\infty)$. $E \subseteq X$ closed.

$$I(E) = \{f \in C(X) : f|_E = 0\}$$

is closed (ideal) in $C(X)$.

Claim: $C(X)/I(E) \simeq C(E)$ isometric isomorphism.

Proof. Note that

$$\begin{array}{ccc}
C(X) & \xrightarrow{R} & C(E) \\
\downarrow Q & \nearrow \tilde{R} & \\
C(X)/I(E) & &
\end{array}$$

where $R(f) = f|_E$, $Qf = \hat{f} = f + I(E)$, $\tilde{R}(Qf) = f|_E = R(f)$. Notice

$$\begin{aligned}
Qf &= Qg \\
\iff f - g &\in I(E) \\
\iff f_E - g_E &= 0 \\
\iff R(f) &= R(g)
\end{aligned}$$

so indeed $\tilde{R}$ is well-defined. Also, $\tilde{R}$ is bijective:

$$\begin{aligned}
\tilde{R}(Qg) &= 0 \\
\iff f|_E &= 0 \\
\iff f &\in I(E) \\
\iff Qf &= 0
\end{aligned}$$

so $\tilde{R}$ is injective.

Surjective by Tietze: given $g \in C(E)$, there exists $f \in C(X)$ such that $f|_E = g$.

For $f \in C(E)$, $g \in I(E)$:

$$\begin{aligned}
\|f|_E\|_\infty &= \sup_{x \in E} |f(x)| \\
&= \sup_{x \in E} |f + g(x)| \\
&\leq \|f + g\|_{C(X)}
\end{aligned}$$

Take inf over $g \in I(E)$:

$$\|\tilde{R}(Qf)\| \leq \|Qf\|_{C(X)/I(E)}$$

So $\tilde{R}$ is a contraction. WLOG, $E \neq X$. Then $C(X)/I(E) \neq \{0\}$. Take $f \in C(X)$ such that

$$f|_E \neq 0$$

For $z \in \mathbb{C}$, let

$$k(z) = \begin{cases} z & , |z| \leq \|f|_E\|_\infty \\ \frac{z}{|z|} \|f|_E\|_\infty & , |z| > \|f|_E\|_\infty \end{cases}$$

Then $k : \mathbb{C} \rightarrow \mathbb{C}$ is continuous. So let $g = k \circ f \in C(X)$. Then

$$\|g\|_\infty = \|k \circ f\|_\infty \leq \|k\|_\infty = \|f|_E\|_\infty$$

For $x \in E$, $|f(x)| \leq \|f|_E\|_\infty$. Therefore $g(x) = k(f(x)) = f(x)$. So $g|_E = f|_E \implies g = f + h, h \in I(E)$. Then

$$\begin{aligned}\|Qf\| &\leq \|g\|_\infty \leq \|f|_E\|_\infty = \|\tilde{R}(Qf)\|_\infty \\ \implies \|\tilde{R}(Qf)\| &= \|Qf\|\end{aligned}$$

□

3.3 Applications of Baire Category Theorem

Definition 3.8. (X, τ) topological space. A subset $A \subseteq X$ is nowhere dense if $\bar{A}$ has empty interior.

Theorem 3.9 (Baire Category Theorem). Let (X, d) be a complete metric space. Then X cannot be written as a countable union of nowhere dense sets.

Equivalently, if $\{U_i\}_{i=1}^\infty$ open dense sets, then

$$\bigcap_{i=1}^\infty U_i$$

is dense.

Definition 3.10 (pointwise bounded). Let X, Y be Banach spaces. $S \subseteq B(X, Y)$ is called pointwise bounded if

$$\forall x, Sx \text{ is bounded in } Y$$

That is, $\forall x \in X, \exists k_x > 0$ such that $\|sx\| \leq k_x, \forall s \in S$.

Theorem 3.11 (Banach Steinhaus Theorem (Principle of Uniform Boundedness)). Let X, Y be Banach spaces, $S \subseteq B(X, Y)$ pointwise bounded, then S is bounded uniformly in $B(X, Y)$. That is,

$$\sup_{s \in S} \|s\| < \infty$$

Proof. $S \subseteq B(X, Y)$. Set $\forall n \in \mathbb{N}$,

$$A_n = \{x \in X : k_x \leq n\}$$

Then A_n is closed. $x_k \in A_n, x_k \rightarrow x$, then

$$\|sx\| = \lim_{k \rightarrow \infty} \|sx_k\| \leq n \implies x \in A_n$$

Then $X = \bigcup_n A_n$, X is complete. By BCT, there exists n_0 such that $(A_{n_0})^\circ \neq \emptyset$. Then there exists $x_0 \in A_{n_0}, r > 0$ such that

$$b_r(x_0) \subseteq A_{n_0}$$

Fix $s \in S, y \in X, \|y\| \leq 1$.

$$sy = \frac{-sx_0 + s(ry + x_0)}{r}$$

Then $x_0, yr + x_0 \subseteq \overline{b_r(x_0)} \subseteq A_{n_0}$. So

$$\|sy\| \leq \frac{\|sx_0\| + \|s(x_0 + ry)\|}{r} \leq \frac{n_0 + n_0}{r} = 2\frac{n_0}{r}.$$

Take $\sup \|y\| \leq 1$, then $\|s\| \leq \frac{2n_0}{r}$, $\forall s \in S$. □

Corollary 3.12. $(T_n)_{n=1}^\infty \subseteq B(X, Y)$. Suppose $\forall x \in X$, $(T_n x)_{n=1}^\infty$ is convergent in Y . Then

$$S = \{T_n : n \in \mathbb{N}\} \text{ is bounded}$$

and there exists $T \in B(X, Y)$ such that

$$Tx = \lim_{n \rightarrow \infty} T_n x$$

and

$$\|T\| \leq \sup_n \|T_n\|$$

Proof. $(T_n x)$ convergent $\forall x$, it is bounded, so S is pointwise bounded, by Banach Steinhaus, S is bounded. Define $T : X \rightarrow Y$ via $Tx = \lim_n T_n x$, linear, and

$$\|Tx\| = \lim_n \|T_n x\| \leq \sup_n \|T_n\| \|x\|.$$

□

Application of Banach Steinhaus: $f \in C(\mathbb{T})$, then $f \rightarrow$ Fourier series $\sum_{n \in \mathbb{N}} \hat{f}(n) e^{int}$.

$$\hat{f}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(e^{it}) e^{-int} dt.$$

Consider the N th partial sum:

$$S_N(f)(e^{it}) = \sum_{n=-N}^N \hat{f}(n) e^{int} \in C(\mathbb{T})$$

Question: Does $S_N f \rightarrow f$ converge pointwisely? uniformly?

Answer: No.

Proof. $S_N f(1) \in \mathbb{C}$, then $\varphi_N : C(\mathbb{T}) \rightarrow \mathbb{C}$ via

$$f \mapsto S_N f(1)$$

is linear.

$$\varphi_N(f) = S_N f(1) = \sum_{n=-N}^N \hat{f}(n) = \frac{1}{2\pi} f(e^{it}) \left(\underbrace{\sum_{n=-N}^N e^{-int}}_{D_N(t)} \right) dt$$

where

$$D_N(t) = \frac{\sin((N + \frac{1}{2})t)}{\sin(\frac{t}{2})}$$

is the Dirichlet's kernel. We know that

$$\|\varphi_N\|_{C(\mathbb{T})^*} = \|D_N\|_{L^1(\mathbb{T})}$$

Fact: $\exists c > 0$ such that

$$\|D_N\|_1 \geq c \log(N)$$

Therefore $\{\varphi_N\}_{N=1}^\infty$ is unbounded in $C(\mathbb{T})^* = B(C(\mathbb{T}), \mathbb{C})$. By B.S., $\{\varphi_N\}_{N=1}^\infty$ is not pointwise bounded. Thus, there exists $f \in C(\mathbb{T})$ such that

$$|\varphi_N(f)| = |S_N f(1)| \rightarrow \infty$$

So, $S_N f \not\rightarrow f$ pointwisely. □

3.4 Open Mapping Theorem

Definition 3.13. We say a map $T \in B(X, Y)$ is open, if $\forall U \subseteq X$ open, we have $T(U) \subseteq Y$ open.

Theorem 3.14. Let $T \in B(X, Y)$, X, Y Banach spaces. If T is surjective, then T is open.

Corollary 3.15 (Banach Isomorphism Theorem). If $T \in B(X, Y)$, and T is surjective, then $T^{-1} \in B(Y, X)$ (continuity of T^{-1} is given for free).

Proof. T is open by open mapping theorem. Then $\forall U \subseteq X$ open, we have

$$(T^{-1})^{-1}(U) = T(U) \text{ is open}$$

Therefore T^{-1} is continuous. □

Corollary 3.16. If $T \in B(X, Y)$ is surjective, then

$$X / \ker T \simeq Y$$

as Banach spaces, via

$$\tilde{T} : X / \ker T \rightarrow Y \quad \hat{x} \mapsto T(x)$$

Proof. $\tilde{T}$ is well-defined bijection:

$$Tx = \tilde{T}(\hat{x}) = \tilde{T}(\hat{y}) = Ty$$

$$\implies T(x - y) = 0 \iff x - y \in \ker T \iff \hat{x} = \hat{y}$$

Also, it is injective, surjective, and linear.

$$\begin{array}{ccc}
X & \xrightarrow{T} & Y \\
Q \downarrow & \nearrow \tilde{T} & \\
X/\ker T & &
\end{array}$$

$$\begin{aligned}
& \|\tilde{T}(\hat{x})\| \\
&= \|T(x)\| \\
&= \|T(x+y)\|, \forall y \in \ker T \\
&\leq \|T\| \cdot \|x+y\|
\end{aligned}$$

Take inf over y :

$$\begin{aligned}
\|\tilde{T}(\hat{x})\| &\leq \|T\| \cdot \|\hat{x}\| \\
\implies \|\tilde{T}\| &\leq \|T\|
\end{aligned}$$

□

Corollary 3.17. If X vector space, then

1. $\|\cdot\|_1, \|\cdot\|_2$ are norms on X , where X is complete with respect to $\|\cdot\|_1, \|\cdot\|_2$.
2. There exists $c > 0$ such that $\|x\|_1 \leq c\|x\|_2, \forall x \in X$. Then there exists $c' > 0$ such that

$$c'\|x\|_2 \leq \|x\|_1 \leq c\|x\|_2, \forall x \in X$$

That is, norms are equivalent.

Proof. $\text{id} : (X, \|\cdot\|_2) \rightarrow (X, \|\cdot\|_1)$, is a bijection, continuous, then id^{-1} is also cts. □

proof of OMT. $Y = TX$. So

$$\begin{aligned}
Y &= T\left(\bigcup_{n=1}^{\infty} nb_1^{\times}(0)\right) \\
&= \bigcup_{n=1}^{\infty} nT(b_1^{\times}(0)) \\
&= \bigcup_n \overline{nT(b_1^{\times}(0))}
\end{aligned}$$

By BCT, there exists n_0 such that $\overline{n_0 T(b_1^{\times}(0))}$ has non-empty interior $\iff \overline{T(b_1^{\times}(0))}$ has nonempty interior. Then there exists $r > 0, y_0 \in Y$ such that

$$b_r(y_0) \subseteq \overline{T(b_1^{\times}(0))}$$

Conversely, if there exists $r > 0$ such that

$$b_r^Y(0) \subseteq \overline{T(b_1^{\times}(0))}$$

Claim: $\overline{Tb_1^{\times}(0)} \subseteq Tb_2^{\times}(0)$. In particular, $b_r^Y(0) \subseteq Tb_2^{\times}(0)$.

Proof. Let $y \in \overline{Tb_1^X(0)}$. Then there exists $x_0 \in b_1^X(0)$ such that

$$\|y - Tx_0\| < \frac{r}{2}$$

Let $y_1 = y - Tx_1 \in b_{\frac{r}{2}}(0) \subseteq \overline{Tb_{\frac{1}{2}}^X(0)}$. Then there exists $x_2 \in b_{\frac{1}{2}}^X(0)$ such that

$$\|y_1 - Tx_2\| = \|y - Tx_1 - Tx_2\| < \frac{r}{4} = \frac{r}{2^2}$$

So $y - Tx_1 - Tx_2 \in b_{\frac{r}{4}}^Y(0) \subseteq \overline{Tb_{\frac{1}{4}}^X(0)}$ Inductively construct $x_k \in b_{\frac{1}{2^{k-1}}}^X(0)$ such that

$$\underbrace{\|y - Tx_1 - Tx_2 - \cdots - Tx_k\|}_{y_k \in b_{\frac{r}{2}}^Y(0)} < \frac{r}{2^k}$$

Then

$$\begin{aligned} \|x\| &\leq \sum_{k=1}^{\infty} \|x_k\| \leq \sum_{k=1}^{\infty} 2^{-k+1} = 2 \\ \implies x &= \sum_{k=1}^{\infty} x_k \in b_2^X(0) \end{aligned}$$

Since T is continuous, we have

$$y = Tx = \sum_{k=1}^{\infty} Tx_k \implies y \in \overline{Tb_1^X(0)} \subseteq Tb_2(0)$$

This is enough to show T is open. Have $b_{\frac{r}{2}}^Y(0) \subseteq Tb_1(0)$. $U \subseteq X$ open, $x \in U$, there exists $\varepsilon > 0$ such that

$$\begin{aligned} x + \varepsilon b_1^X(0) &\subseteq U \\ \implies Tx + \varepsilon Tb_1^X(0) &\subseteq T(U) \\ \implies Tx + \varepsilon b_{\frac{r}{2}}(0) &\subseteq T(U) \end{aligned}$$

Therefore, $T(U)$ is open. □

□

Corollary 3.18. $X = \mathbb{F}^n$. Any two norms on X are equivalent and X is complete.

Proof. Let $\|\cdot\|_1$ be any norm, and $\|\cdot\|_2$ be the Euclidean norm. e_i be the standard basis for X . Then $x = \sum_{i=1}^n x_i e_i$.

$$\begin{aligned} \|x\|_1 &\leq \sum_i |x_i| \|e_i\|_1 \\ &\leq \|x\|_2 \underbrace{\left(\sum_{i=1}^n \|e_i\|_1^2 \right)^{\frac{1}{2}}}_C \\ &= C \|x\|_2 \end{aligned} \quad \text{C.S.}$$

Therefore, $T = \text{id} : (X, \|\cdot\|_2) \rightarrow (X, \|\cdot\|_1)$ is a continuous bijection. If we know $(X, \|\cdot\|_1)$ was complete, we get $\|\cdot\|_2, \|\cdot\|_1$ are equivalent by the last corollary. Let $S := \{x \in X : \|\cdot\|_2 = 1\}$ be the usual sphere, it is compact in $\|\cdot\|_2$ sense. Then T is continuous implies TS is compact in $(X, \|\cdot\|_1)$. Then

$$\inf_{x \in S} \|Tx\| = r > 0$$

Claim: $T^{-1}b_r^{\|\cdot\|_1}(0) \subseteq b_1^{\|\cdot\|_2}(0)$.

Proof. Suppose $\|x\|_1 < r$ with $\|x\|_2 = t \geq 1$, then

$$\left\| \frac{x}{t} \right\|_2 = 1, \frac{x}{t} \in S$$

but $\frac{x}{t} \in b_r(0)$, $\left\| \frac{x}{t} \right\| < r < \text{dist}(0, TS)$, contradiction. □

Therefore $rT^{-1}(b_1)(b_1^{\|\cdot\|_1}(0)) \subseteq b_1^{\|\cdot\|_2}(0)$, then

$$\|rT^{-1}\| \leq 1 \implies \|T^{-1}\| \leq \frac{1}{r}$$

□

Definition 3.19. X, Y Banach space, then the direct sum is

$$X \bigoplus Y = \{(x, y) : x \in X, y \in Y\}$$

componentwise addition, scalar multiplication. Therefore, $X \oplus Y$ is a vector space.

We can define norms for $1 \leq p < \infty$:

$$X \bigoplus_p Y : \|(x, y)\|_p = (\|x\|^p + \|y\|^p)^{\frac{1}{p}}$$

So in particular, $\|(x, y)\|_1 = \|x\| + \|y\|$.

$$X \bigoplus_\infty Y : \|(x, y)\|_\infty = \max\{\|x\|, \|y\|\}.$$

Notice that $\|\cdot\|_p$ are all equivalent. Generally, just consider $X \oplus Y$. Let $D \subseteq X$ subspace, linear map $T : D \rightarrow Y$.

Graph of T :

$$\mathcal{D}(T) = \{(x, Tx) : x \in D\}$$

We say T is closed if $\mathcal{D}(T)$ is a closed subset of $X \oplus Y$.

Example 3.20. $T \in B(X, Y)$, then T is closed.

$X = Y = (C[0, 1], \|\cdot\|_\infty)$, $D = C^1([0, 1])$, $T = \frac{d}{dx}$, then T is closed.

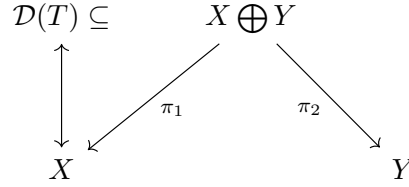
$f_n \in D$ such that $f_n \rightarrow f$, then $Tf_n \rightarrow g$, $(f_n, Tf_n) \rightarrow (f, g)$.

Theorem 3.21 (Closed Graph Theorem). X, Y Banach. $T : X \rightarrow Y$ is closed $\iff T \in B(X, Y)$.

Proof. We have $\pi_1(x, y) = x$, $\pi_2(x, y) = y$, then π_1, π_2 are continuous and linear. Notice $\pi_1|_{\mathcal{D}(T)} : \mathcal{D}(T) \rightarrow X$ is continuous and bijective. By Banach Isomorphism Theorem, we have

$$(\pi_1|_{\mathcal{D}(T)})^{-1} : X \rightarrow \mathcal{D}(T) \text{ bounded}$$

Therefore $T = \pi_2 \circ (\pi_1|_{\mathcal{D}(T)})^{-1}$ is a composition of bounded maps, hence bounded.



□

Corollary 3.22. Let $T : X \rightarrow Y$. Then $T \in B(X, Y) \iff \forall x_n \rightarrow 0, Tx_n \rightarrow y \in Y \implies y = 0$.

Proof. $(\implies)$ is automatic: T is continuous then T is continuous at 0.

$(\impliedby)$ $\mathcal{D}(T)$ is closed. $x_n \rightarrow x \implies x_n - x \rightarrow 0$. Then $Tx_n \rightarrow y \implies T(x_n - x) \rightarrow y - Tx$. By assumption, $y - Tx = 0 \implies Tx = y$. □

3.5 Hahn-Banach Theorem

Theorem 3.23 (HB Theorem 1). Let X be a Banach space (or normed linear space). Let $M \subseteq X$ be a linear subspace.

$$\varphi_0 : M \rightarrow \mathbb{F} \text{ is linear and bounded on } M$$

That is, $\|\varphi_0\|_{M^*} < \infty$. Then there exists a norm preserving extension $\varphi \in X^*$. That is, there exists $\varphi \in X^*$ such that

1. $\varphi|_M = \varphi_0$,
2. $\|\varphi\|_{X^*} = \|\varphi_0\|_{M^*}$.

Corollary 3.24 (1). Given $0 \neq x \in X$. Then there exists $\varphi \in X^*$, $\|\varphi\| = 1$ such that

$$\varphi(x) = \|x\|$$

In particular,

1. $\|x\|_X = \sup\{|\varphi(x)| : \varphi \in X^*, \|\varphi\| = 1\}$,
2. X^* separates points of X .

Proof. $x \neq 0$, $M = \mathbb{F}x$, define

$$\varphi_0(\lambda x) = \lambda\|x\|, \lambda \in \mathbb{F}$$

Then we obtain $\|\varphi_0\| = 1$. By HB Theorem 1, there exists $\varphi \in X^*$, $\|\varphi\| = 1$, extending φ_0 .

$$\implies \varphi(x) = \|x\|$$

Also, $x \neq y \implies x - y \neq 0 \implies$ there exists $\varphi \in X^*$ such that

$$\varphi(x - y) = \|x - y\| \neq 0 \implies \varphi(x) \neq \varphi(y)$$

$i(x) = \hat{x}$, where

$$X \xrightarrow{i} X^{**}$$

$x \in X, \varphi \in X^*, \hat{x}(\varphi) = \varphi(x)$. Then

$$\begin{aligned} \|x\| &= \varphi(x) \\ &= |\varphi(x)| \\ &\leq \sup_{\psi \in X^*, \|\psi\|=1} |\psi(x)| \\ &\leq \|x\| \end{aligned}$$

which implies (1). □

Corollary 3.25.

$$X \xhookrightarrow{i} X^{**}$$

is isometric.

$$\begin{aligned} \|\hat{x}\| &= \sup_{\|\varphi\|=1} |\hat{x}(\varphi)| \\ &= \sup_{\|\varphi\|=1} |\varphi(x)| \\ &= \|x\| \end{aligned}$$

For any normed space X , X^{**} is Banach space.

$$\overline{X} = \overline{i(X)} \subseteq X^{**}$$

Example 3.26. $X = C_0, X = l_1, X^{**} = l_1^* = l_\infty$.

Now, we define the following for the set up for proof of HB.

Definition 3.27. Let X be a Banach space.

$$p : X \rightarrow \mathbb{R}$$

is called a sublinear functional if $\forall t > 0, p(tx) = t(p(x)), \forall x \in X$, and $p(x, y) \leq p(x) + p(y)$.

Example 3.28. 1. $p(x) = \|x\|$.

2. X Banach space. Let $x \in X$ be fixed, $p_x : B(X) \rightarrow \mathbb{R}$ via $p_x(T) = \|Tx\|$.

3. $\varphi \in X^*, x \in X, T \in B(X)$. Then

$$p_{\varphi, x}(T) = |\varphi(Tx)|.$$

Theorem 3.29. Let X be a Banach space over $\mathbb{R}$, $p : X \rightarrow \mathbb{R}$ sublinear functional, $M_0 \subseteq X$ linear subspace, $f_0 : M_0 \rightarrow \mathbb{R}$ linear such that $f_0(x) \leq p(x), \forall x \in M_0$. Then there exists $f : X \rightarrow \mathbb{R}$ linear such that

1. $f|_{M_0} = f_0$,
2. $f(x) \leq p(x), \forall x \in X$.

3.6 Weak Topology

$(X, \|\cdot\|)$ normed vector space. $(X^*, \|\cdot\|_{X^*})$ dual Banach space. Take $Y \subseteq X^*$ linear subspace (possibly not closed) that separates points of X . Then we can define LCTCSs:

1.

$$(X, \tau_Y), \tau_Y = \text{LC topology coming from } \{|\varphi| : \varphi \in Y\}$$

We say τ_Y is the weak topology on X associated to Y . When $Y = X^*$, we just call τ_X the weak topology on X .

2. (X^*, τ_{X^*}) weak x -topology on X^* . Recall that

$$X \hookrightarrow X^{**} \text{ isometrically, } x \mapsto \hat{x}, \hat{x}(\varphi) = \varphi(x)$$

Therefore $X \subseteq (X^*)^*$, we can view X as “ Y ” here. Clearly X separates points of X^* :

$$\text{Proof. } \varphi \neq \psi \in X^* \iff \exists x \in X, \varphi(x) \neq \psi(x) \iff \hat{x}(\varphi) \neq \hat{x}(\psi). \quad \square$$

$$x_\lambda \in X, x_\lambda \rightarrow x \in X \text{ in } \tau_Y, Y \subseteq X^* \iff \varphi(x_\lambda) \rightarrow \varphi(x), \forall \varphi \in Y. \text{ Similarly, } \varphi_\lambda \in X^*, \varphi_\lambda \xrightarrow{\tau_X} \varphi \iff \varphi_\lambda(x) \rightarrow \varphi(x), \forall x \in X.$$

Note that we could take $Y \subseteq X^*$ a subset that separates points. Then

$$\begin{aligned} x_\lambda \xrightarrow{\tau_Y} x &\iff |\varphi(x_\lambda - x)| \rightarrow 0, \forall \varphi \in Y \\ &\iff |\psi(x_\lambda - x)| \rightarrow 0, \forall \psi \in \text{span}(Y) \end{aligned}$$

Therefore, $\tau_Y = \tau_{\text{span}(Y)}$.

Now, let X, Y normed vector space, then what does the dual look like for $(B(X, Y), \|\cdot\|_{B(X, Y)})$?

Definition 3.30 (Weak operator topology (WOT) on $B(X, Y)$). This is induced by functionals:

$$f_{x,y} \in B(X, Y)^*, x \in X, y \in Y$$

$$f_{x,y}(T) = \varphi(Tx), T \in B(X, Y)$$

Then it separates points by Hahn-Banach theorem.

Proof. $T \neq S \implies \exists x \in X$ such that $Tx \neq Sx$ in Y . By HB, there exists $\varphi \in Y^*$ such that $\varphi(Tx) \neq \varphi(Sx)$ $\square$

Then $T_\lambda \xrightarrow{WOT} T \iff \varphi(T_\lambda x) \rightarrow \varphi(Tx), \forall \varphi \in Y^*, x \in X$. If $X = Y = H$ Hilbert space, $\langle \cdot, \cdot \rangle : H \times H \rightarrow \mathbb{F}$. By Riesz-Representation Theorem, if $\varphi \in H^*, \exists! y \in H$ such that

$$\varphi(\xi) = \langle \xi, y \rangle, \forall \xi \in H$$

So $T_\lambda \xrightarrow{WOT} T \in B(H) \iff \langle T_\lambda \xi, y \rangle \rightarrow \langle T\xi, y \rangle, \forall \xi, y \in H$.

Strong Operator Topology on $B(X, Y) \iff$ topology of point-norm convergence.

$$T_\lambda \in B(X, Y)$$

$$T_\lambda \xrightarrow{SOT} T \in B(X, Y)$$

$$\iff T_\lambda x \xrightarrow{\|\cdot\|_Y} Tx, \forall x \in X$$

$$\iff \|T_\lambda x - Tx\| \rightarrow 0, \forall x \in X$$

Therefore norm convergence $\implies$ SOT convergence $\implies$ WOT convergence.

SOT seminorms:

$$\mathcal{P} = (P_x)_{x \in X}, P_x(T) = \|Tx\|$$

Let X be a LCTVS, then

$\tau_{\mathcal{P}}$ -topology induced by seminorms $P \in \mathcal{P}$

Theorem 3.31. Let $\varphi : X \rightarrow \mathbb{F}$. TFAE:

1. φ is continuous
2. φ is continuous at 0
3. $\ker \varphi$ is closed
4. $\exists P_1, \dots, P_n \in \mathcal{P}, \alpha_i > 0$ such that

$$|\varphi(x)| \leq \sum_{i=1}^n \alpha_i P_i(x), \forall x \in X$$

Proof. (1) $\implies$ (2): immediate.

(2) $\implies$ (3): let $x_\lambda \in \ker \varphi$,

$$x_\lambda \xrightarrow{\tau_R} x \in X, x_\lambda - x \rightarrow 0$$

Then $\varphi(x) = \varphi(x - x_\lambda), \forall \lambda$

$$\implies x \in \ker \varphi$$

(4) $\implies$ (1): If $x_\lambda \xrightarrow{\tau_P} x \iff$

$$p(x_\lambda - x) \rightarrow 0, \forall p \in \mathcal{P}$$

$$\implies \sum_{i=1}^n \alpha_i p_i(x_\lambda - x) \rightarrow 0$$

Therefore $|\varphi(x_\lambda - x)| \rightarrow 0$.

(3) $\implies$ (4) : $\varphi \neq 0$. There exists $x_0 \in X$ such that $\varphi(x_0) = 1$. $x_0 \notin \ker \varphi$, which is closed by assumption.

$$\implies F \subseteq \mathcal{P} \text{ finite, } r > 0 \text{ such that } x_0 + U_{F,r} \subseteq (\ker \varphi)^C$$

$$x \in U_{F,r} \iff p(x) < r, \forall p \in F.$$

$$\varphi(\underbrace{x_0 + U_{F,r}}_{\neq 0}) = 1 + \varphi(U_{F,r})$$

$$\implies -1 \notin \varphi(U_{F,r})$$

$x \in U_{F,r}, |\lambda| \leq 1$. Then

$$\lambda x \in U_{F,r}, \forall |\lambda| \leq 1$$

Claim: $\varphi(U_{F,r}) \in \mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$.

Claim: $|\varphi(x)| \leq \frac{1}{r} \sum_{p \in F} p(x)$.

□

3.7 Geometric Hahn-Banach theorems

Hahn-Banach is about separation of nice subsets.

Example 3.32. $(X, \|\cdot\|)$ normed space, $x \neq y \in X$, by HB, there exists $\varphi \in X^*$ such that $\varphi(x) \neq \varphi(y)$.

If $Y \subsetneq X$ closed subspace,

$$\{0\} \neq (X/Y)^* \simeq Y^\perp = \{\varphi \in X^* : \varphi(y) = 0, \forall y \in Y\}$$

Therefore, we can separate Y from $x \in X \setminus Y$ by continuous functions.

GOAL: Given LCTVS $(X, \tau_{\mathcal{P}})$ and convex sets $A, B \subseteq X$ such that $A \cap B = \emptyset$. We want $\varphi \in (X, \tau_{\mathcal{P}})^* := X^*$ continuous linear functional such that

$$\varphi(A) \cap \varphi(B) = \emptyset$$

Notice $\varphi(A), \varphi(B) \subseteq \mathbb{C}$ are convex. In this case, $\varphi(A), \varphi(B)$ will be separated by a line in $\mathbb{C}$. Replace φ by $\lambda\varphi, \lambda \in \mathbb{T}$ so that the line is vertical, $\varphi \mapsto \lambda\varphi, \varphi(A) \mapsto (\lambda\varphi)(A)$. In X , we have

$$A \subseteq \{x \in X : \operatorname{Re}\varphi(x) \leq \alpha\}$$

$$B \subseteq \{x \in X : \operatorname{Re}\varphi(x) \geq \alpha\}$$

Aim: $A, B \subseteq X$ convex disjoint, there exists $\alpha \in \mathbb{R}, \varphi \in (X, \tau_{\mathcal{P}})^*$ such that

$$\operatorname{Re}\varphi(x) < \alpha \leq \operatorname{Re}\varphi(y), \forall x \in A, y \in B$$

Example 3.33. (l^1, τ_{c_0}) , weak *-topology.

$$A = \{x = (x_i)_{i=1}^\infty \in l^1 : \sum_{i=1}^\infty x_i = 0\}$$

$B = \{\delta_1\}$. Then A, B convex disjoint,

$$\{\delta_i\}_{i=1}^\infty \text{ indicator function of } i \in \mathbb{N}$$

Then $(l_1, \tau_{c_0})^* = c_0$.

Claim: $\forall \varphi \in c_0, \varphi(A) \cap \varphi(B) = \emptyset$.

Proof. Take $0 \neq \varphi = (a_1, a_2, \dots) \in c_0$. Then there exists $m \in \mathbb{N}$ such that $a_m \neq 0$. Consider

$$(\delta_m - \delta_n)_{n=1}^\infty \subseteq A$$

$$\implies \varphi(\delta_m - \delta_n) = a_m - a_n \rightarrow a_m \neq 0, n \rightarrow \infty$$

Therefore A is not contained in $\ker \varphi$.

$$\implies \varphi(A) = \mathbb{C}, \forall \varphi \neq 0.$$

Therefore we cannot separate $\varphi(A)$ from $\varphi(B)$. □

What is the issue here? A is not closed, not open.

Theorem 3.34 (Separation Theorem I). Let $(X, \tau_{\mathcal{P}})$ be LCTVS, $A, B \subseteq X$ disjoint convex. If A is open, then there exists $\varphi \in (X, \tau_{\mathcal{P}})^*$ and $\alpha \in \mathbb{R}$ such that

$$\operatorname{Re}\varphi(x) < \alpha \leq \operatorname{Re}\varphi(y), \forall x \in A, y \in B$$

There are three steps:

1. find an appropriate sublinear functional on X ,
2. do an appropriate extension,
3. get continuity “for free”.

Minkowski functionals

Definition 3.35 (Minkowski functional associated to U). Let X be as above, $0 \in U$ convex and open. Then

$$P_U(x) = \inf\{t > 0 : x \in tU\} \in [0, \infty)$$

Theorem 3.36. $P_U : X \rightarrow [0, \infty)$ is a sublinear functional and

$$U = \{x \in X : P_U(x) < 1\}.$$

Proof. $x \in sU, t > 0 \iff tx \in tsU$.

$$\begin{aligned} P_U(tx) &= \inf\{r > 0 : tx \in rU\} \\ &= \inf\{ts : tx \in tsU\} \\ &= t \inf\{s : x \in sU\} \\ &= tP_U(x) \end{aligned}$$

If $x \in sU, y \in tU, s, t > 0$, then

$$x + y = \underbrace{\left(\frac{s}{s+t}\left(\frac{x}{s}\right) + \frac{t}{s+t}\left(\frac{y}{t}\right)\right)}_{\in U \text{ by convexity}}(s+t)$$

Therefore $x + y \in (s+t)U$, hence

$$\begin{aligned} P_U(x+y) &\leq s+t \\ \implies P_U(x+y) &\leq P_U(x) + P_U(y) \end{aligned}$$

Therefore P_U is a sublinear functional.

$x \in U$ open, $t \mapsto tx$ is continuous, then there exists $\delta > 0$ such that $(1+\delta)x \in U$. Hence

$$\begin{aligned} x &\in \frac{1}{1+\delta}U \\ \implies P_U(x) &< 1 \end{aligned}$$

If $P_U(x) < 1$, then there exists $\lambda \in (0, 1)$ such that $x \in \lambda U$.

$$x = \lambda \frac{x}{\lambda} + (1-\lambda)(0) \in U$$

□

Let X be a vector space over $\mathbb{F}$, $\varphi : X \rightarrow \mathbb{F}$ linear. Write $\langle x, \varphi \rangle := \varphi(x)$, $\langle \cdot, \cdot \rangle$ bilinear form. For $Y \subseteq X^*$ subspace. $(X, \|\cdot\|)$ normed space, $\sigma(X, Y)$ -topology: “weak topology on X ”. The continuous functions are

$$\{x \mapsto \langle x, \varphi \rangle : \varphi \in Y\}$$

τ_Y -weak topology = $\sigma(X, Y)$ top. weak topology on X^* . $\sigma(X^*, X)$ topology. Then the continuous functions are

$$\{\varphi \mapsto \langle x, \varphi \rangle : x \in X\}$$

Definition 3.37. Let X be a LCTVS. $A \subseteq X$ convex. A face of A is $\emptyset \neq F \subseteq A$ convex, and whenever $x, y \in A, t \in (0, 1)$, and $(1-t)x + ty \in F$, we have $x, y \in F$.

Remark 3.38. If $F = \{z\}$, we call z an extreme point of A . That is, we cannot write z as a non-trivial convex combination. We use $\text{Ext}(A)$ to denote the set of extreme points of A .

Example 3.39. The sides of a triangle are faces, and the three angles are extreme points.

Example 3.40. $X = (L^1[0, 1], dx)$. $\overline{b_1(x)}$ =unit ball. $f \in L^1[0, 1], \|f\|_1 = 1$

$$\int_0^1 |f(x)| dx = 1$$

$$\implies \exists t_0 \in (0, 1) \text{ such that } \int_0^{t_0} |f(x)| dx = \int_{t_0}^1 |f(x)| dx = \frac{1}{2}$$

Let $f_1 = \frac{2f\chi_{[0, t_0]} + 2f\chi_{[t_0, 1]}}{2}$, so f_1 is not a extreme point. Therefore, $\overline{b_1(L^1([0, 1]))}$ has no extreme points.

Theorem 3.41 (Krein-Milman Theorem). Let $\emptyset \neq K \subseteq (X, \tau)$, where (X, τ) is a LCTVS, and K is convex and compact. Then $K = \overline{\text{conv}(\text{Ext}(K))}$.

Corollary 3.42 (Krein-Milman). If $(X, \|\cdot\|)$ normed vector space, then

$$\overline{b_1(X^*)}$$

is the weak*-closed convex hull of its extreme points.

Proof. Analogue $\implies K = \overline{b_1(X^*)}$ is weak* compact convex. By KMT, we get what we want. $\square$

Example 3.43. $L^1[0, 1]$ is not a dual space. $l^1 = (C_0)^*$. More generally, X locally compact, Hausdorff, $C_0(X)^* = M(X)$ =complex Borel measures. $\text{Ext}(\overline{b_1(M(X))}) = \{\delta_x : x \in X\}$.

$$\int_x f d\delta_x = f(x)$$

KMT. Fix $\emptyset \subseteq K \subseteq X$, K compact and convex. Firstly, we want to show $\text{Ext}(K) \neq \emptyset$:

Proof. We would use Zorn's lemma here. Fix a face $\emptyset \neq F_0 \subseteq K$. Let

$$\Lambda = \{F \subseteq F_0 : F \text{ is a closed face of } F_0\}$$

we can give a partial order by reverse inclusion: $F_1 \leq F_2$ if $F_2 \subseteq F_1$. Let $\mathcal{C} = \{F_\alpha\}_\alpha$ chain in Λ . An upper bound for $\mathcal{C}$:

$$F = \bigcap_{\alpha} F_\alpha \neq \emptyset \text{ by compactness}$$

Also, it is closed and convex, and it is a face of F_0 , as $x, y \in F_0, (1-t)x+ty \in F \implies (1-t)x+ty \in F_\alpha, \forall \alpha \implies x, y \in F_\alpha, \forall \alpha$. By Zorn's lemma, there exists a maximal element of the partial order set, that is, there exists a minimal closed face $F \subseteq F_0$. Fix a continuous linear functional $\varphi \in (X, \tau)^*$. Let

$$\alpha_{\varphi, F} = \inf_{x \in F} (\operatorname{Re} \varphi(x)) = \min_{x \in F} \operatorname{Re} \varphi(x)$$

$\operatorname{Re} \varphi$ continuous, F compact, therefore

$$F_\varphi = \{x \in F : \operatorname{Re} \varphi(x) = \alpha_{\varphi, F}\}$$

Then F_α is closed, convex, and is a face of F , as $\operatorname{Re} \varphi(tx + (1-t)y) = t\operatorname{Re} \varphi(x) + (1-t)\operatorname{Re} \varphi(y)$ is minimum means that each term must be at minimum. By minimality, $F_\varphi = F$. If $x \neq y \in F$, then there exists $\varphi \in (X, \tau)^*$ such that

$$\operatorname{Re} \varphi(x) \neq \operatorname{Re} \varphi(y).$$

But both evaluate to $\alpha_{\varphi, F}$, therefore $(X, \tau)^*$ does not separate points. Therefore, $\operatorname{Ext}(K) \cap F_0 \neq \emptyset, \forall F_0 \subseteq K$ closed face. $\square$

Secondly, we want to show that suppose there exists $x_0 \in K \setminus \underbrace{\operatorname{conv}(\operatorname{Ext}(K))}_B$. Consider an open convex set $x_0 \in A$ such that $A \cap B = \emptyset$. Find $\varphi \in (X, \tau)^*$ such that $t \in \mathbb{R}$,

$$\operatorname{Re} \varphi(x_0) \in \operatorname{Re} \varphi(A) < t \leq \operatorname{Re} \varphi(B)$$

Therefore $\alpha_{\varphi, F} = \min_{x \in F} \operatorname{Re} \varphi(x) < t \leq \operatorname{Re} \varphi(B)$. Consider

$$F_\varphi = \{x \in K : \operatorname{Re} \varphi(x) = \alpha_{\varphi, K}\}$$

Then $F_\varphi \cap B = \emptyset$. Therefore $F_\varphi \cap \operatorname{Ext}(K) = \emptyset$, contradicts to step 1. $\square$

X compact Hausdorff space. $(C(X), \|\cdot\|_\infty)$. Then the dual space

$$C(X)^* = M(X) = \{\mu \text{ complex Borel measure on } X\}$$

and

$$\mu(f) = \int_X f d\mu, f \in C(X)$$

Every $\mu \in M(X)$, has a polar decomposition

$$d\mu = h d|\mu|, |\mu| = \text{total variation } (\geq 0), h \text{ is Borel measurable, } |h(x)| = 1, \forall x \in X.$$

$$\implies \mu(f) = \int_X f d\mu = \int_X f h d|\mu|$$

So

$$\|\mu\|_{M(X)} = |\mu|(X) = \sup_{\|f\|_\infty \leq 1} |\mu(f)|$$

$$P(X) = \{\mu \geq 0 \text{ in } M(X) \text{ such that } \mu(X) = 1\} = \text{probability measures}$$

$$P(X) \subseteq \overline{b_1(M(X))}, \text{ and}$$

$$P(X) = \text{weak}^* \text{ compact convex subset of } (M(X), \text{weak}^*)$$

Proposition 3.44. $\text{Ext}(P(X)) = \{\delta_x : x \in X\}$, δ_x Dirac measures, $\delta_x(f) = f(x)$.

Proof. Let $\mu \in \text{Ext}(P(X))$. Fix $0 \leq f \leq 1$ in $C(X)$. Set $\lambda = \mu(f) \in [0, 1]$. If $\lambda \in (0, 1)$, then $\forall g \in C(X)$,

$$\mu_1(g) = \frac{1}{\lambda} \mu(fg) = \frac{1}{\lambda} \int_X fg d\mu$$

$$\mu_2(g) = \frac{1}{1-\lambda} \mu((1-f)g) = \frac{1}{1-\lambda} \int_X g(1-f) d\mu$$

Then $\mu_1, \mu_2 \in P(X)$, and

$$\mu = \lambda \mu_1 + (1-\lambda) \mu_2$$

Since $\mu \in \text{Ext}(P(X))$, we have $\mu = \mu_1 = \mu_2$. Therefore $\forall g \in C(X)$,

$$\mu(g) = \mu_1(g) = \frac{1}{\lambda} \mu_1(fg) = \frac{\mu_1(fg)}{\mu(f)}$$

$$\implies \mu(fg) = \mu(g)\mu(f)$$

Also, if $\mu(f) = 0$, we get

$$\begin{aligned} |\mu(fg)| &\leq \mu(|fg|) \\ &= \mu(f|g|) \\ &= \int_X f|g| d\mu \\ &\leq \|g\|_\infty \int_X f d\mu \\ &= \|g\|_\infty \cdot \mu(f) = 0 \end{aligned}$$

Therefore $\mu(fg) = 0 = \mu(f)\mu(g)$.

If $\mu(f) = 1, 0 \leq f \leq 1, 0 \leq 1 - f \leq 1$, and $\mu(1 - f) = 0$. Therefore $\mu((1 - f)g) = 0, \forall g \in C(X)$ as in the last case. Thus

$$\begin{aligned}\mu(g) - \mu(fg) &= 0 \\ \implies \mu(fg) &= \mu(g) = \mu(f)\mu(g)\end{aligned}$$

Therefore, in all cases, $\mu(fg) = \mu(f)\mu(g), \forall g \in C(X), 0 \leq f \leq 1, g \in C(X)$.

$$C(X) = \text{span}_{\mathbb{R}}\{f \in C(X) : 0 \leq f \leq 1\}$$

Therefore μ is a multiplicative and linear functional.

Claim: $\exists x \in X$ such that $\ker \delta_x \supseteq \ker \mu$.

Proof. X is compact, $1 \in C(X), \forall x \in X, \exists f_x \in C(X)$,

$f_x(x) \neq 0$, then $\{y : f_x(y) \neq 0\}$ is an open cover of X

X is compact, so there exists $x_1, \dots, x_n \in X$ such that

$$X = \bigcup_{i=1}^n \underbrace{\{y : f_{x_i}(y) \neq 0\}}_{\text{open}}$$

Therefore we get

$$\begin{aligned}0 < f &= \sum_{i=1}^n |f_i|^2 \in C(X) \\ \implies \frac{1}{f} &\in C(X) \\ \implies 1 &= \mu(1) = \mu\left(\frac{f}{f}\right) = \mu(f)\mu\left(\frac{1}{f}\right) = \mu\left(\frac{1}{f}\right)\mu\left(\sum_{i=1}^n f_{x_i}\bar{f}_{x_i}\right) = 0\end{aligned}$$

as $\sum_i \mu(f_{x_i})\mu(\bar{f}_{x_i}) = 0$.

Suppose next, given $\delta_x \in P(X)$. If $\delta_x = \lambda\mu + (1 - \lambda)\nu, \mu, \nu \in P(X)$. Then

$$\begin{aligned}|f(x)| &= |\delta_x(f)| \\ &= \delta_x(|f|) \\ &= \lambda\mu(|f|) + (1 - \lambda)\nu(|f|) \\ &\geq \lambda\mu(|f|) \\ &\geq |\lambda\mu(f)|\end{aligned}$$

Therefore $\ker(\mu) \subseteq \ker \delta_x$. □

Now by claim, we have $g \in C(X), g - \mu(f) \cdot 1 \in \ker(\mu) \implies g \in \ker \delta_x$,

$$\begin{aligned}\implies \delta_x(g - \mu(g)1) &= 0 \\ \implies g(x) - \mu(g) &= 0\end{aligned}$$

Therefore $\mu = \delta_x$, as desired. □

4 Adjoint Operator

X, Y Banach spaces, $T \in B(X, Y)$. The adjoint of T is $T^* \in B(Y^*, X^*)$ given by

$$\langle x, T^* \varphi \rangle = \langle Tx, \varphi \rangle, x \in X, \varphi \in Y^*$$

Theorem 4.1. $T \in B(X, Y)$,

1. $\|T\| = \|T^*\|$,
2. $B(X, Y) \rightarrow B(Y^*, X^*), T \mapsto T^*$ is linear,
3. $(TS)^* = S^*T^*, T \in B(Y, Z), S \in B(X, Y)$,
4. T^* is weak *- weak *- continuous, $T^* : Y^* \rightarrow X^*$,
5. $\text{id}_X : X \rightarrow X, (\text{id}_X)^* = \text{id}_{X^*}$,
6. $T^{**} \in B(X^{**}, Y^{**})$ satisfies

$$T^{**}|_X = T.$$

Remark 4.2. If $X = \mathbb{C}^n$ with e_i basis, $Y = \mathbb{C}^m$ with f_i basis. Then $(e_i^*), (f_i^*)$ are dual basis for X^*, Y^* .

$$e_i^*(e_j) = \delta_{i,j}, f_i^*(f_j) = \delta_{i,j}.$$

Therefore

$$\begin{aligned} T \in B(X, Y) &\iff A = [a_{ij}] \in M_{m,n}(\mathbb{C}) \\ T^* \in B(Y^*, X^*) &\iff B = A^T \end{aligned}$$

where

$$\begin{aligned} a_{ij} &= \langle Te_j, f_i^* \rangle \\ &= \langle e_j, T^* f_i^* \rangle \\ &= \langle T^* f_i^*, e_j \rangle \end{aligned}$$

Proof. 1. We have

$$\begin{aligned} \|T^*\| &= \sup_{\varphi \in Y^*, \|\varphi\| \leq 1} \|T^* \varphi\|_{X^*} \\ &= \sup_{\|\varphi\| \leq 1} \sup_{x \in X, \|x\| \leq 1} |\langle x, T^* \varphi \rangle| \\ &= \sup_{\|x\| \leq 1} \sup_{\|\varphi\| \leq 1} |\langle Tx, \varphi \rangle| \\ &= \sup_{\|x\| \leq 1} \|Tx\| \\ &= \|T\| \end{aligned}$$

2. direct
3. We have

$$\begin{aligned}
& \langle x, (TS)^* \varphi \rangle \\
&= \langle TSx, \varphi \rangle \\
&= \langle Sx, T^* \varphi \rangle \\
&= \langle x, S^* T^* \varphi \rangle
\end{aligned}$$

4. Suppose $\varphi_\lambda \in Y^*$ via $\varphi_\lambda \rightarrow \varphi$ (weak *), that is,

$$\forall y \in Y, \langle y, \varphi_\lambda \rangle \rightarrow \langle y, \varphi \rangle$$

Then for all $x \in X$,

$$\langle x, T^* \varphi_\lambda \rangle = \langle Tx, \varphi_\lambda \rangle$$

we get $T^* \varphi_\lambda \rightarrow T^* \varphi$ weak * in X^* . Therefore we obtain the weak * - weak * - continuity.

5. direct
6. We have

$$\begin{aligned}
X &\xrightarrow{i_X} X^{**}, x \mapsto \hat{x} \\
Y &\xrightarrow{i_Y} Y^{**}, y \mapsto \hat{y}
\end{aligned}$$

WANT: $Y^{**} \ni T^{**} \hat{x} = \widehat{Tx}$. Take $\varphi \in T^*$,

$$\begin{aligned}
\langle \varphi, T^{**} \hat{x} \rangle &= \langle T^* \varphi, \hat{x} \rangle \\
&= \langle x, T^* \varphi \rangle \\
&= \langle Tx, \varphi \rangle \\
&= \langle \varphi, \widehat{Tx} \rangle \\
\implies T^{**} \hat{x} &= \widehat{Tx}
\end{aligned}$$

□

Theorem 4.3. Let $S \in B(Y^*, X^*)$. Then there exists $T \in B(X, Y)$ such that $T^* = S$ if and only if S is weak *-weak *- continuous.

Theorem 4.4. ($\implies$) part (4) of the previous theorem.

($\impliedby$) Consider $S^* \in B(X^{**}, Y^{**})$. Consider

$$T := S^*|_X \in B(X, Y^{**})$$

Fix $x \in X$, $S^* \hat{x} \in Y^{**}$.

Claim: $S^* \hat{x} \in Y$.

Proof. Let $\varphi_\lambda \rightarrow \varphi$ weak * , where both of them $\in Y^*$. Therefore

$$\langle \varphi_\lambda, S^* \hat{x} \rangle = \langle S \varphi_\lambda, \hat{x} \rangle \rightarrow \langle S \varphi, \hat{x} \rangle = \langle \varphi, S^* \hat{x} \rangle.$$

$$\implies S^* \hat{x} \text{ is a weak } ^* \text{ continuous function on } (Y^*, \tau_Y)^*$$

$$\implies S^* \hat{x} \in Y \subseteq Y^{**}$$

□